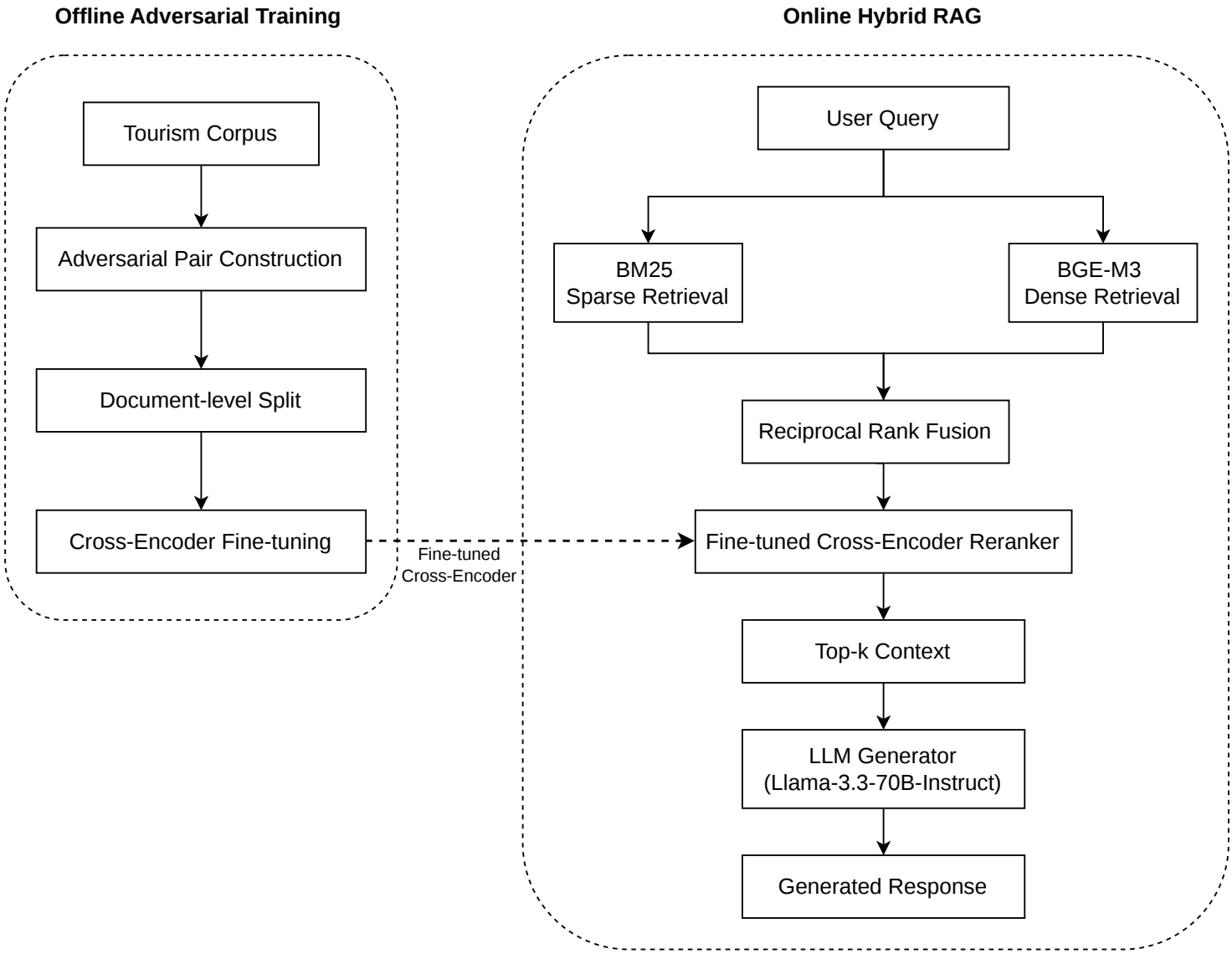
Offline Adversarial Training
Tourism Corpus
Adversarial Pair Construction
Document-level Split
Cross-Encoder Fine-tuning
Fine-tuned
Cross-Encoder
Online Hybrid RAG
User Query
BM25
Sparse Retrieval
BGE-M3
Dense Retrieval
Reciprocal Rank Fusion
Fine-tuned Cross-Encoder Reranker
Top-k Context
LLM Generator
(Llama-3.3-70B-Instruct)
Generated Response